

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Intermediate Filament Organization (GO:0	0.24532001	50	2.010e-09	1.083e-05	KRT35:3 KRT8:8 KRT17:23 KRT4:107 KRT74:135 KRT25:138 KRT27:152
Defense Response To Symbiont (GO:0140546	-0.15364760	102	8.624e-08	2.323e-04	RNASEL:17 IKBKE:40 IFNLR1:72 LYST:94 FADD:185 CD207:362 MYD88:386
Positive Regulation Of Cytokine Production	-0.08797580	274	6.052e-07	9.841e-04	CD14:8 TYK2:27 TLRA:37 IKBKE:40 IL23R:110 IL23A:111 IL2RB:140
Sensory Perception Of Stimuli (GO:0007608)	0.16662938	72	6.638e-07	8.941e-04	OR10J5:161 OR5B21:197 UBR3:209 OR51B5:354 OR2D2:383 OR13A:395 OR51B4:443
Cellular Response To Cytokine Stimulus (+	-0.08659643	233	5.721e-06	6.164e-03	OSMR:11 TYK2:27 HAS2:36 TLRA:37 IL2RB:42 IRAK3:46 IFNAR1:105
Positive Regulation Of Cytokine-Mediated	-0.21645893	36	7.041e-06	6.323e-03	IKBKE:40 TRIM44:85 TICAM2:146 FADD:185 USP27X:629 IL7:830 RIPK1:1101
Positive Regulation Of Leukocyte Mediate	-0.24726151	27	8.757e-06	6.741e-03	CD160:53 IL23R:110 IL23A:111 IL2RB:140 FADD:185 IL23A:111 IL2RB:140
Regulation Of T Cell Mediated Cytotoxici	-0.34990085	13	1.255e-05	8.451e-03	IL23R:110 IL23A:111 IL2RB:140 FADD:185 IL2A:532 HLA-DRA:1768 IL2B:2459
Defense Response To Virus (GO:0051607)	-0.10477969	139	2.085e-04	1.248e-02	RNASEL:17 IKBKE:40 IFNLR1:72 LYST:94 FADD:185 CD207:362 MYD88:386
Detection Of Chemical Stimulus Involved	0.16445414	54	2.951e-05	1.590e-02	OR10J5:161 OR2D2:383 OR13C4:395 OR10A7:509 OR2C1:857 OR10K2:873 OR13F:1958
Detection Of Chemical Stimulus Involved	0.15758634	55	5.356e-05	2.624e-02	OR10J5:161 OR2D2:383 OR13C4:395 OR10A7:509 OR2C1:857 OR10K2:873 OR13F:1958
Regulation Of Interleukin-17 Production	-0.20535856	32	5.846e-05	2.625e-02	TYK2:27 TLRA:37 IL23R:110 IL23A:111 FOXF3:227 MYD88:386 LY9:444
Cytokine-Mediated Signaling Pathway (GO:	-0.08122927	197	8.912e-05	3.694e-02	OSMR:11 TYK2:27 TLRA:37 IL2RB:42 IRAK3:46 IL23R:110 CCL20:131 IL2RB:140
Sulfation (GO:0051923)	0.29846354	14	1.105e-04	4.254e-02	SULT1C2:982 SULT1A3:1669 SULT1C3:1677 SULT1A2:1690 SULT1A4:1774 SULT1B1:1809 SULT1C4:1878
DNA Repair (GO:0006281)	-0.06888869	276	1.406e-04	5.051e-02	MEN1:51 DCLRE1A:89 RAD18:99 ERCC5:226 BABAM1:230 NPM1:278 RAD9A:294
Cholesterol Biosynthetic Process (GO:000	0.22191334	24	1.684e-04	5.065e-02	HMGCS1:221 HSD17B7:254 G6PD:299 MSMO1:307 TM7SF2:613 FDF1:673 ACLY:709
Positive Regulation Of T Cell Mediated I	-0.23168617	22	1.692e-04	5.056e-02	IL23R:110 IL23A:111 IL2RB:140 FADD:185 IL2A:532 TRAF2:728 TRAF6:1277
Supramolecular Fiber Organization (GO:00	0.06575162	283	1.516e-04	5.065e-02	COL12A1:2 KRT35:3 KRT85:8 KRT17:23 COL1A2:51 COL11A2:52 COL12A2:1
Regulation Of Type II Interferon Product	-0.12529073	74	1.973e-04	5.594e-02	CD14:8 TYK2:27 TLRA:37 IL23R:110 IL23A:111 IL2RB:140 FADD:185
Secondary Alcohol Biosynthetic Process (+	0.22031601	23	2.555e-04	6.872e-02	HMGCS1:221 HSD17B7:254 G6PD:299 MSMO1:307 TM7SF2:613 FDF1:673 ACLY:709
Sterol Biosynthetic Process (GO:0016126)	0.20270096	27	2.676e-04	6.872e-02	HMGCS1:221 HSD17B7:254 G6PD:299 MSMO1:307 TM7SF2:613 FDF1:673 ACLY:709
B Cell Mediated Immunity (GO:0019724)	-0.07778602	14	3.199e-04	7.354e-02	IL2RB:42 FCGR2B:484 CD70:569 MS2H:1634 CD19:142 L12R:2216 IL9R:2546
Epithelial Cell Differentiation (GO:0030	0.09777983	113	3.379e-04	7.354e-02	KRT35:3 KRT17:23 KRT4:107 KRT25:138 KRT74:135 KRT25:138 KRT27:152
Positive Regulation Of Type II Interfero	-0.14327663	52	3.549e-04	7.354e-02	CD14:8 TYK2:27 TLRA:37 IL23R:110 IL23A:111 IL2RB:140 FADD:185
Purine Ribonucleotide Biosynthetic Proce	-0.14307800	52	3.216e-04	7.354e-02	COASY:35 AMPD3:44 PAPSS1:168 ADCY5:213 ADCY3:356 SLC25A13:64 ADCY5:599
Regulation Of Transferrase Activity (GO:0	-0.04939614	19	3.494e-04	7.354e-02	PPPIR3F:457 PARP4:481 KAT7B:889 PINX1:1083 SLAMF8:1591 LDB1:1593 JMDJB:2415
Immunoglobulin Mediated Immune Response	-0.21151093	23	4.472e-04	9.824e-02	IL2RB:42 FCGR2B:484 GAP7:526 GNAO1:878 HLA-DRA:1768 CD19:1942 HLA-DPM:1926
Gland Development (GO:0048732)	-0.12355800	66	5.243e-04	9.834e-02	EZR:137 FADD:185 PTFA1:221 IL2A:532 SLC9A3R1:275 HES1:361 MSN:1059
Positive Regulation Of Natural Killer Ce	-0.27761287	13	5.293e-04	9.834e-02	CD160:53 KLRK1:189 IL23A:111 IL2A:532 IL2B:2459 CADM1:2631 SLAMF6:3012
Positive Regulation Of Inflammatory Resp	-0.10554200	89	5.885e-04	1.057e-01	OSMR:11 TLRA:37 IL23A:111 MGST2:130 NFKBIB:405 NFKBIA:467 IL21:503
Regulation Of Inflammatory Response (GO:	-0.06866746	211	6.126e-04	1.057e-01	TLRA:37 IL23A:111 MACR1:116 MGST2:130 LRRC19:144 KLRK1:189 CXCL17:182
Sensory Perception Of Chemical Stimulus	0.18675707	28	6.276e-04	1.057e-01	OR10J5:161 OR5B21:197 UBR3:209 OR51B5:354 OR2D2:383 OR51B4:443 CALHM3:444
T Cell Activation (GO:0042110)	-0.10635378	86	6.621e-04	1.061e-01	NLR3:148 DPP4:211 SLAMF9:225 FOXF3:227 TNFSF14:272 TNFSF14:272 TNFSF14:272
Branchched-Chain Amino Acid Catabolic Proc	0.23833254	17	6.698e-04	1.061e-01	BCKDHA:28 BCKDK:558 ECHS1:1072 MCC61:370 HMGCL:1333 HMGCL:1333 HMGCL:1333
Cellular Response To cAMP (GO:0071320)	-0.17826431	30	7.297e-04	1.123e-01	EZR:137 SLC5A5:177 CFTR:292 SLC26A5:170 AKAP9:580 RABGEF1:621 HCN1:661
B Cell Activation Involved In Immune Res	-0.26862857	13	7.985e-04	1.171e-01	LIG4:356 FCGR2B:484 GAP7:526 MS2H:1634 RNF8:1664 MSH5:1904 CD180:1937
Primary Alcohol Catabolic Process (GO:00	0.36576399	7	8.044e-04	1.171e-01	HAO1:336 SULT1A3:1669 SULT1A2:1690 SULT1B1:1809 SULT1A7:1686 SULT1A1:2685
Positive Regulation Of T Cell Mediated C	-0.24809659	15	8.796e-04	1.247e-01	IL23R:110 IL23A:111 IL2RB:140 FADD:185 IL2A:532 HLA-DRA:1768 IL2B:2459
Interstrand Cross-Link Repair (GO:003629	-0.16165128	35	9.391e-04	1.297e-01	DCLRE1A:89 RFWD3:458 ERCC1:1624 RNF8:1664 POLN:1847 DCLRE1C:1850
3'-Phosphoadenosine 5'-Phosphosulfate Me	0.25453878	14	9.764e-04	1.315e-01	PAPSS1:168 SULT1A3:1669 SULT1C3:1677 SULT1A2:1690 SULT1B1:1809 SULT1C4:1878

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_FORMATION_OF_THE_CORNFIED_ENVE	0.22988006	78	2.293e-12	1.488e-08	KRT35:3 KRT8:8 KRT17:23 KRT4:107 CSTA:108 KRT74:135 KRT25:138
REACTOME_KERATINIZATION	0.21463823	85	8.124e-12	2.635e-08	KRT35:3 KRT8:8 KRT17:23 KRT4:107 CSTA:108 KRT74:135 KRT25:138
REACTOME_OLFACTORY_SIGNALING_PATHWAY	0.14941932	122	1.235e-08	2.377e-05	OR10J5:161 OR5B21:197 OR4C3:201 OR1A4:256 OR51B5:354 ADCY3:356 OR2D2:383
KEGG_OLFACTORY_TRANSDUCTION	0.14085380	136	1.466e-08	2.377e-05	GUCA1C:33 OR10J5:161 OR5B21:197 OR4C3:201 OR1A4:256 CALML5:328 OR51B5:354
WP_NETWORK_MAP_OF_SARSCOV2_SIGNALING_PAT	-0.11568989	182	7.649e-08	9.926e-05	CD14:8 MMP25:77 CRP:104 CCL20:131 JUN:345 VP53:338 MYD88:386
DODD_NYOPHARYNGEAL_CARCIOMA_DM	-0.04216611	1245	7.542e-07	8.155e-04	CD14:8 POP1:10 FLAD1:59 HOXC9:88 FKBP14:93 RAD18:99 VANGL2:124
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	-0.05753161	627	1.007e-06	9.931e-04	OSMR:11 RNASEL:17 TYK2:27 PPM1B:31 IL2RB:42 IL7:830 IL2RB:140
REACTOME_SENSORY_PERCEPTION	0.07886407	319	1.367e-06	1.108e-03	GUCA1C:33 GRXCRI:69 ESPNL:100 METAP1:106 OR10J5:161 OR5B21:197 OR4C3:201
WALLACE_PROSTATE_CANCER_RACE_UP	-0.09049910	237	1.669e-06	1.203e-03	SAMSN1:103 RGS16:139 HERC6:155 UBE2L:6168 GZMA:175 STAG3:249 ANPEP:458
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	-0.09493614	208	2.458e-06	1.595e-03	OSMR:11 IL2RB:42 IFNLR1:72 IL7RB:86 IFNAR1:105 IL23R:110 IL23A:111
REACTOME_INTERFERON_ALPHA_BETA_SIGNALING	-0.20170955	43	4.749e-06	2.766e-03	RNASEL:17 TYK2:27 IFNAR1:105 IFR8:665 MX2:719 IRF3:792 SOC1S:1835
WP_MEASLES_VIRUS_INFECTION	-0.12595818	110	5.116e-06	2.766e-03	TYK2:27 TLRA:37 IKBKE:40 IL2RB:42 IFNAR1:105 FADD:185 CCND1:255
WP_TCELL_ACTIVATION_SARSCOV2	-0.13940706	85	8.988e-06	4.486e-03	TYK2:27 IFNAR1:105 IL23R:110 IL23A:111 IL2RB:140 GRAP2:192 CCND1:255
PID_IL12_2PATHWAY	-0.16417096	58	1.539e-05	7.132e-03	TYK2:27 IL2RB:42 IL2RB:140 GZMA:175 IL2RB:140 IL2RB:408 IL2RB:506 IL2A:532
LIU_OVARIAN_CANCER_TUMORS_AND_XENOGRAFTS	-0.03455559	1341	2.747e-05	1.188e-02	DCHS2:4 CD14:8 OSMR:11 CDC20B:20 HAS2:36 TLRA:37 HEG1:38
BROWNE_INTERFERON_RESPONSIVE_GENES	-0.16345102	53	3.871e-05	1.570e-02	UBE2L:6168 PDZD2:287 MYD88:386 TRIM21:619 THEMIS2:643 IFI44L:721 MK2:779
WP_SULFATION_BIOTRANSFORMATION_REACTION	0.30276653	15	4.908e-05	1.769e-02	PAPSS1:168 G6PD:299 SULT1C2:982 SULT1C3:1677 SULT1A2:1690 SULT1A4:1774 SULT1B1:1809
BIOCARTA_TH1TH2_PATHWAY	-0.26974184	19	4.692e-05	1.769e-02	IL2RB:140 IL2RB:140 IL2RB:140 IL2RB:140 IL2RB:140 IL2RB:140 IL2RB:140 IL2RB:140
REACTOME_METABOLISM_OF_LIPIDS	0.05442713	685	5.617e-05	1.822e-02	FABP4:32 ACBD5:61 MTMR1:91 PEMT:136 TNFAIP8L:1165 PRKCD2:170 CERSE6:202
POOLA_INVASIVE_BREAST_CANCER_UP	-0.07747521	229	5.537e-05	1.822e-02	SAMSN1:103 RGS16:139 GZMA:175 CALU:171 MBP1B:181 KIF20A:193 STAG2:287
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	-0.07009192	268	8.115e-05	2.375e-02	L23A:111 CCL20:131 UBE2L:6168 SLC5A5:177 CYP11A2:673 TNFSF14:272 PDZD2:287
BLANCO_MELO_CHRONCHIAL_EPITHELIAL_CELLS_I	-0.04910406	540	1.025e-04	2.375e-02	LINS1:29 TLL11:34 IKBKE:40 DLX2:49 TCF7:6077:74 IL23A:111 RGS16:139
VERHAAK_AML_WITH_NPM1_MUTATED_UP	-0.09185065	151	9.982e-05	2.375e-02	CD14:8 TLRA:37 IKBKE:40 IFNAR1:105 IL2RB:140 FADD:185 JUN:345
SMID_BREAST_CANCER_NORMAL_LIKE_UP	-0.05684414	408	8.647e-05	2.375e-02	GDF7:1 STK10:52 GZMA:175 IL2D2:288 DPP4:211 ARHGAP4:215 C3:232
ICHIBA_GRAFT_VERSUS_HOST_DISEASE_D7_UP	-0.11882130	91	9.038e-05	2.375e-02	CD14:8 NFKBIB:405 IL2RB:140 MARCO:441 CRYAA:462 NFKBIA:467 PIK3R1:471 CSK:546
REACTOME_TCR_SIGNALING	-0.10893433	108	9.303e-05	2.375e-02	CD101:166 GRAP2:192 FBXW11:379 UBE2D1:454 NFKBIA:467 PIK3R1:471 CSK:546
WP_INFLAMMATORY_TCR_DISEASE_SIGNALING	-0.17173048	43	9.798e-05	2.375e-02	TLRA:37 IL23R:110 IL23A:111 IL2RB:140 FADD:185 JUN:345 IL21:503
WP_CHOLESTEROL_SYNTHESIS_DISORDERS	0.26496889	18	9.947e-05	2.375e-02	HMGCS1:221 MSMO1:307 FDP5:515 FDF1:673 CYP51A1:856 MVD:1484 DHCRT4:1979
KEGG_RIG_I_LIKE_RECEPTOR_SIGNALING_PATHW	-0.15438268	52	1.181e-04	2.642e-02	IKBKE:40 FADD:185 NFKBIB:359 NFKBIA:467 IL2A:532 TRAF2:728 TRIM25:741
ROETH_TERT_TARGETS_UP	-0.29589836	14	1.264e-04	2.733e-02	HEG1:38 HERC6:155 PLAG2:44A:271 IFI44L:721 MX1:946 NRP1:1173 IFIT3:1437
PYEON_CANCER_HEAD_AND_NECK_VS_CERVICAL_U	-0.08475195	169	1.470e-04	3.076e-02	CALML4:68 DDX59:159 FGFR1OP:268 FMRI:301 CEP44:302 UBXN7:332 RFC5:339
WP_TH17_CELL_DIFFERENTIATION_PATHWAY	-0.13287565	68	1.522e-04	3.086e-02	TYK2:27 IL23R:110 IL23A:111 FOXF3:227 JUN:345 NFKBIA:467 IL21:503
WP_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWAY	-0.11855328	85	1.598e-04	3.142e-02	CD14:8 TLRA:37 IKBKE:40 IFNAR1:105 TICAM2:146 FADD:185 JUN:345
REACTOME_CHOLESTEROL_BIOSYNTHESIS	0.21717179	25	1.710e-04	3.263e-02	HMGCS1:221 HSD17B7:254 MSMO1:307 FDP5:515 TM7SF2:613 FDF1:673 CYP51A1:856
WP_NOVEL_INTRACELLULAR_COMPONENTS_OF_RIG	-0.14626799	54	2.017e-04	3.635e-02	IKBKE:40 FADD:185 NFKBIB:359 NFKBIA:467 TRAF2:728 TRIM25:741 IRF3:792
KEGG_TOLL_LIKE_RECEPTOR_SIGNALING_PATHWA	-0.11760519	84	1.962e-04	3.635e-02	CD14:8 TLRA:37 IKBKE:40 IFNAR1:105 TICAM2:146 FADD:185 JUN:345
WP_IMMUNE_RESPONSE_TO_TUBERCULOSIS	-0.22710205	21	3.150e-04	5.241e-02	TYK2:27 IFNAR1:105 SOC1S:1835 MX1:946 IFIT3:1437 STAT2:2318 IRF9:2659
ALTEMEIER_RESPONSE_TO_LPS_WITH_MECHANICA	-0.10402589	101	3.064e-04	5.241e-02	CD14:8 OSMR:11 CCL20:131 ADM:218 BCL3:308 CLEC4D:323 CCRL2:338
PID_IL23_PATHWAY	-0.16712834	35	3.116e-04	5.241e-02	TYK2:27 IL23R:110 IL23A:111 IL2RB:140 IL2RB:140 NFKBIA:467 PIK3R1:471
WP_TYPE_I_INTERFERON_INDUCTION_AND_SIGNA	-0.19860865	27	3.545e-04	5.751e-02	TYK2:27 TLRA:37 IKBKE:40 IFNAR1:105 MYD88:386 IRF3:792 TLR7:1122

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	genevals
Influenza	-0.07989551	438	1.250e-08	1.226e-04	TLRA:37 IRAK3:46 ITGAE:79 PLS1:81 CRP:104 IFNAR1:105 IL23A:111
Encephalitis, St. Louis	-0.12417661	95	2.952e-05	4.597e-02	CD14:8 TYK2:27 IKBKE:40 CRP:104 DNM1:128 C322 FCGR2B:484
Allergic rhinitis (disorder)	-0.10609096	129	3.280e-05	4.597e-02	CD14:8 TLRA:37 IL2RB:42 IFNLR1:72 IL23R:110 FOXF3:227 PGM3:383
Autoimmune Disorders	-0.04467636	835	1.523e-05	4.597e-02	IGFALS:7 CD14:8 TYK2:27 TLRA:37 IL2RB:42 IL2RB:42 IFNLR1:72
Bacterial Infections	-0.08561739	211	1.931e-05	4.597e-02	CD14:8 TLRA:37 IRAK3:46 CRP:104 IL23A:111 FADD:185 C3:232
Decreased antibody level in blood	-0.17364036	50	2.186e-05	4.597e-02	CLCA4:252 CDND1:255 CFTR:292 PIK3R1:471 IL21:503 HLAS:781 CARD11:819
Pneumonia	-0.06023665	411	3.149e-05	4.597e-02	CD14:8 IL2RB:33 TLRA:37 IL2RB:42 IRAK3:46 LAMC2:65 ITGAE:79
Hypogammaglobulinemia	-0.13424019	78	4.232e-05	5.919e-02	LAMC2:65 IL2RB:140 FOXF3:227 CLCA4:252 CDND1:255 CFTR:292 PIK3R1:471
Primary Sj?gren's syndrome	-0.11084483	112	5.213e-05	5.683e-02	CD14:8 IKBKE:40 IL23R:110 IL23A:111 IL2RB:142 GZMA:175 RAPH1:256
Crohn Disease	-0.04299771	711	1.130e-04	1.108e-01	CD14:8 OSMR:11 TYK2:27 TLRA:37 IRAK3:46 CD160:53 ITGAE:79
Arthritis, Psoriatic	-0.06865287	159	1.631e-04	1.173e-01	IFNLR1:72 CRP:104 IL23R:110 IL23A:111 IL2RB:140 DPP4:211 CCND1:255
Immunologic Deficiency Syndromes	-0.04838540	541	1.356e-04	1.173e-01	IGFALS:7 TYK2:27 IL2RB:33 TLRA:37 IL2RB:42 FADD:185 LYST:94
Infection	-0.05554469	396	1.634e-04	1.173e-01	CD14:8 RNASEL:17 TLRA:37 IKBKE:40 CYREN:100 CRP:104 IFNAR1:105
Ulcerative Colitis	-0.04175088	717	1.674e-04	1.173e-01	CD14:8 OSMR:11 TYK2:27 TLRA:37 IL2RB:42 CD160:53 LAMC2:65
airway disease	-0.12216907	78	1.944e-04	1.271e-01	HAS2:36 TLRA:37 IL2RB:42 EZR:137 TASR2B:167 C3:232 CFTR:292
IGA Glomerulonephritis	-0.07729534	191	2.398e-04	1.442e-01	CD14:8 TLRA:37 IL23R:110 EZR:137 ADM:218 C3:232 MYD88:386
Increased bilirubin level (finding)	0.21602227	24	2.498e-04	1.442e-01	ADK:29 SPTB:316 SLC25A13:364 SLCA4:148 UGT1A6:721 KIF23: